

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Short Answer Text(High)

Assessment Tool Weightage: 30%

Dr

QUESTIONS		Total Marks: 50	
Q.No	Question	Bloom's Level	Marks
1	UPS Product Tracking Information System UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type	BL1 – Remember	5

	(e.g, flight, truck), and a deliveryRoute. Create an Entity Relationship diagram that captures this information about the UPS system.		
2	Course Management System Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as • course with attributes name, department, and c-number • section with attributes s-number and enrollment, and dependent as a weak entity set on course • room with attributes r-number, capacity, and building Show an E-R diagram illustrating the use of all three additional entity sets listed.	BL2 – Understand	5
3	University Registrar's Office Management System A university registrar's office maintains data about the following entities: (a) courses, including number, title, credits, syllabus, and prerequisites; (b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; (c) students, including student-id, name, and program; (d) instructors, including identification number, name, department, and title. (e) Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office.	BL2 – Understand	5

4	Hospital Management System Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. Construct appropriate tables for the above ER Diagram : Patient(SS#, name, insurance) Physician (name, specialization) Test-log(SS#, test-name, date, time) Doctor-patient (physician-name, SS#) Patient-history(SS#, test-name, date)	BL2 – Understand	5
5	Company Project Management System Construct an ER Diagram for Company having following details : Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations. □ A department controls a number of PROJECT. Projects have a unique name, number and a single location. □ Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee. □ Employee's DEPENDENT are tracked for health insurance purposes (dependentname, birthdate, relationship to employee).	BL2 – Understand	5
6	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember	5
7	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand	5

8	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand	5
9	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand	5
10	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember	5

Code of Conduct:

I E. Rohan - 192524482 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: E.Rohan

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1)

Ans: The UPS Product Tracking Information System stores information about shipped items, retail centers, and transportation events. The main entities and their relationships are as follows.

Entities and Attributes

1. Shipped_Item

- Item_Number (Primary Key)
- Weight
- Dimensions
- Insurance_Amount
- Destination
- Final_Delivery_Date

2. Retail_Center

- Unique_ID (Primary Key)
- Type
- Address

3. Transportation_Event

- Schedule_Number (Primary Key)
- Type (Flight/Truck)
- Delivery_Route

Relationships

- A **Retail Center** receives many **Shipped Items**, but each shipped item is received at only one retail center.
 - Relationship: **Receives** (1 : M)
- A **Shipped Item** travels through one or more **Transportation Events**, and each transportation event can carry many shipped items.
 - Relationship: **Transported_By** (M : N)

2)

Ans: The university exam scheduling system can be modeled using four entities: **Course, Section, Room, and Exam**. The **Section** entity is a weak entity because it depends on the **Course** entity for identification.

Entities and Attributes

1. Course

- Name (Primary Key)
- Department
- Course_Number

2. Section (Weak Entity)

- Section_Number (Partial Key)
- Enrollment

3. Room

- Room_Number (Primary Key)
- Capacity
- Building

4. Exam

- Exam_ID (Primary Key)
- Time

Relationships

- One **Course** has many **Sections**.
- Each **Section** belongs to one **Course**.
- Each **Section** has one **Exam**.
- Each **Exam** is conducted in one **Room**.
- A **Room** can be used for many exams at different times.

3)

Ans: The University Registrar's Office maintains information about students, courses, instructors, course offerings, and student enrollments.

Entities and Attributes

1. Course

- Course_Number (Primary Key)
- Title
- Credits
- Syllabus
- Prerequisites

2. Course_Offering

- Offering_ID (Primary Key)
- Year
- Semester
- Section_Number
- Timings
- Classroom

3. Student

- Student_ID (Primary Key)
- Name
- Program

4. Instructor

- Instructor_ID (Primary Key)
- Name
- Department
- Title

Relationships

- One **Course** can have many **Course Offerings**.
- Each **Course Offering** is taught by one or more **Instructors**.
- A **Student** can enroll in many **Course Offerings**.
- • Each **Course Offering** can have many **Students**.
- • The **Enrollment** relationship stores the student's **Grade**.

4)

Ans: A Hospital Management System maintains records of patients, physicians, medical tests, and patient history. The system also records which physician treats which patient.

Entities and Attributes

1. Patient

- SS# (Primary Key)
- Name
- Insurance

2. Physician

- Name (Primary Key)
- Specialization

3. Test_Log

- SS# (Foreign Key)
- Test_Name
- Date
- Time

Relationships

- A **Physician** treats many **Patients**, and a patient can be treated by multiple physicians. (M:N)
- A **Patient** undergoes multiple tests. (1:M)

5)

Ans: The Company Project Management System stores information about departments, employees, projects, and employee dependents. It also records project assignments and department management.

Entities and Attributes

1. Department

- Department_Name (PK)
- Locations

2. Employee

- SSN (PK)
- Name
- Address
- Salary
- Sex
- Birth_Date

3. Project

- Project_Number (PK)
- Project_Name
- Location

4. Dependent

- Dependent_Name
- Birth_Date
- Relationship

Relationships

- One department is managed by one employee.
- One department controls many projects.
- One department has many employees.
- Employees work on multiple projects (M:N).
- Employees can have multiple dependents.
- An employee may supervise other employees.

6)

Ans: **Schema**

A **schema** is the logical structure or blueprint of a database. It defines the tables, attributes, relationships, and constraints of the database. The schema is designed once and changes very rarely.

Example:

Student(Student_ID, Name, Department, Age)

This table structure is the schema.

Instance

An **instance** is the actual data stored in the database at a particular point in time. The instance changes whenever records are inserted, updated, or deleted.

Difference Between Schema & Instance

Schema	Instance
Blueprint of the database	Actual data stored in the database
Changes rarely	Changes frequently
Defines structure	Contains records
Created during database design	Updated during database operation

Types of Schema

Schema Type	Description
Logical Schema	Defines the logical structure of the database such as tables, attributes, and relationships.
Physical Schema	Describes how data is physically stored on storage devices, including files and indexes.
View Schema	Represents the portion of the database visible to a particular user or application.

7)

Ans:

A Bank Management System stores information about customers, accounts, loans, branches, and employees. It helps manage customer accounts, banking transactions, and loan details efficiently.

Entities and Attributes

1. Customer

- Customer_ID (Primary Key)
- Name
- Address
- Phone_Number

2. Account

- Account_Number (Primary Key)
- Account_Type
- Balance

3. Loan

- Loan_Number (Primary Key)
- Loan_Type
- Loan_Amount

4. Branch

- Branch_ID (Primary Key)
- Branch_Name
- Location

5. Employee

- Employee_ID (Primary Key)
- Name
- Designation

Relationships

- A **Customer** can own one or more **Accounts**. (1:M)
- A **Customer** can apply for multiple **Loans**. (1:M)
- A **Branch** manages many **Accounts** and **Loans**. (1:M)
- A **Branch** employs many **Employees**. (1:M)

8)

Ans: The **Extended Entity Relationship (EER) Model** is an advanced version of the Entity Relationship (ER) model. It extends the basic ER model by introducing additional concepts such as generalization, specialization, and aggregation.

Generalization is the process of combining two or more lower-level entities that share common attributes into a single higher-level entity called a **superclass**. This helps reduce data redundancy by storing common attributes only once. For example, the entities **Student** and **Teacher** can be generalized into a superclass called **Person**, as both share common details like Name and Address.

Specialization is the opposite of generalization. It divides a higher-level entity into multiple lower-level entities based on their specific characteristics. Each subclass inherits the attributes of the superclass and can also have its own unique attributes. For example, an **Employee** entity can be specialized into **Manager**, **Engineer**, and **Accountant**, where each subclass has additional role-specific information.

Aggregation is used when a relationship between two entities needs to be treated as a separate entity so that it can participate in another relationship. It is useful for representing complex relationships in a database. For example, the relationship **Works_On** between an Employee and a Project can be aggregated so that a Manager can monitor that work assignment.

Conclusion:

In conclusion, the **EER model** provides a more powerful and flexible approach to database design than the traditional ER model. It improves data organization, reduces redundancy, and accurately represents complex relationships, making it suitable for designing large and real-world database systems.

9)

Ans: A Hospital Management System can be designed using the **Extended ER (EER) Model** by applying **inheritance** and **weak entities**.

Superclass

Person

- Person_ID (PK)
- Name
- Address
- Phone_Number

Subclasses (Inheritance)

Patient

- Patient_ID
- Insurance_Details

Doctor

- Doctor_ID
- Specialization
- Qualification

Weak Entity

Medical_Record

- Record_Number (Partial Key)
- Diagnosis
- Treatment
- Visit_Date

The **Medical_Record** is a weak entity because it cannot exist without a Patient.

Relationships

- A **Patient** has many **Medical Records**. (1:M)

10)

Ans:

A **Database Administrator (DBA)** is responsible for managing, maintaining, and securing an organization's database system. The DBA ensures that the database operates efficiently, securely, and is always available to authorized users.

Responsibilities of a DBA

1. Database Installation and Configuration

- Installs DBMS software and configures databases according to organizational requirements.

2. Database Security

- Creates user accounts, assigns permissions, and protects data from unauthorized access.

3. Backup and Recovery

- Performs regular backups and restores data in case of hardware failure or accidental data loss.

4. Performance Monitoring

- Optimizes database performance by tuning queries, managing indexes, and monitoring system resources.

5. Data Integrity

- Ensures that data remains accurate, consistent, and reliable by enforcing integrity constraints.

6. Database Maintenance

- Updates software, applies security patches, and manages storage to ensure smooth operation.

7. Troubleshooting

- Identifies and resolves database errors, connectivity issues, and performance bottlenecks.

Conclusion

The Database Administrator plays a vital role in ensuring the security, performance, and availability of the database. A skilled DBA helps maintain reliable database operations and protects valuable organizational data.